Constants: (Elementary charges)

(C) Coulomb $= 6.25\times10^{18}$ e's

$k = 9\times10^{9}\ N\cdot m^2/C^2$

$\epsilon_0 = 8.85\times10^{-12}\ C^2/N\cdot m^2$ (Permitivity constant)

$c = 3\times10^{8}$ m/s (light speed)

$e = 1.6\times10^{-19}$ C

$m_p =$ [redacted] $\times10^{-27}$ kg (Proton)

$m_e = 9.11\times10^{-31}$ kg (Electron)

1 | Current and Resistance

Electron Current:

$Ne = ie\,\Delta t$ | $ie = Ne\cdot A\cdot V_d$

where $\boxed{Ne}$ is the # of electrons that flow through the cross section during Δt and $\boxed{ie}$ is the rate of the electron flow (Electrons per second)

where $\boxed{A}$ is the cross sectional area (πr^2) and $\boxed{V_d}$ is the Drift Speed

Conventional Current:

$I = e\cdot ie,\ I = \frac{Q}{\Delta t}$

Amp = Coulomb/sec

Current Density:

$I = J\cdot A$ where J = Current density and A = area

$J = I/A$

$J = \frac{Ne\cdot e^2\cdot \tau}{m}\cdot\vec{E}$

$J = \sigma\cdot E$

Conductivity:

$\sigma = \frac{Ne\cdot e^2\cdot\tau}{m}$

where m is the mass of the electron

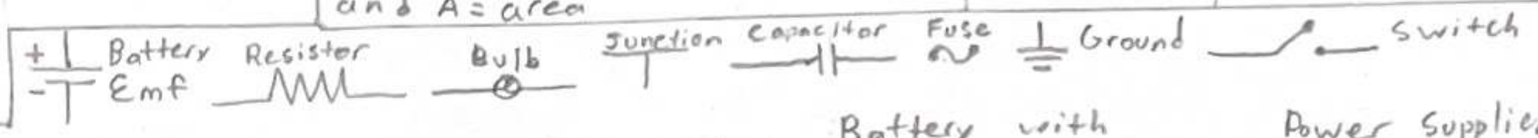
Resistivity: Ω

$\rho = \frac{1}{\sigma} = \frac{m}{Ne\cdot e^2\cdot\tau}$

$E = \frac{J}{\sigma} = \frac{\Delta V}{Length}$

Wire Resistance:

$R = \frac{\rho\cdot L}{A}$



2 | Circuits

Ohm's Law:

$V = IR$

Power (Watts):

$P = IV = \frac{Energy}{\Delta t}$

$= I^2R$

$= \frac{V^2}{R}$

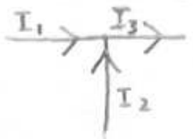
(Loop Law):

$\mathcal{E} - IR = 0$

$\therefore \mathcal{E} = IR$

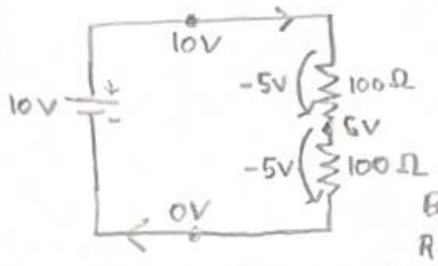
Kirchoff's Laws:

① Junction Law: $\sum I_{in} = \sum I_{out}$



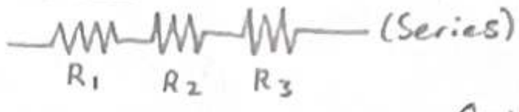
$I_1 + I_2 = I_3$

② Voltage Law: $\sum_{Loop} V_i = 0$

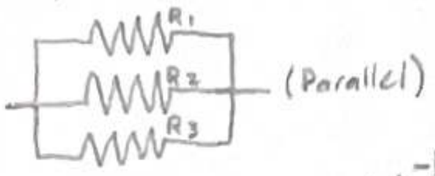


Battery (source)
Resistor (load)

Resistance:

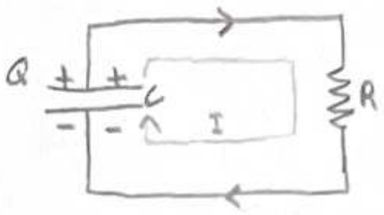


$R_{eq} = R_1 + R_2 + R_3 \ldots + R_n$



$R_{eq} = \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \ldots + \frac{1}{R_n}\right)^{-1}$

RC Circuit:



$Q = Q_0 e^{-t/\tau}$

where $\tau = RC$ is the time constant

Battery with internal Resistance:

$\Delta V_r = \mathcal{E} - Ir$

Ammeter:

- Measures current
- Must be placed in series

Voltmeter:

- Measures potential difference
- Must be placed in parallel

Power supplied by a battery:

$P = \mathcal{E}I$

Resistors lose energy at a rate of:

$P_r = IV = I^2R = \frac{V^2}{R}$

Potential Difference

$\Delta V_r = IR$

$\Delta V_r = \mathcal{E}$

3 | Electric Charge/Forces

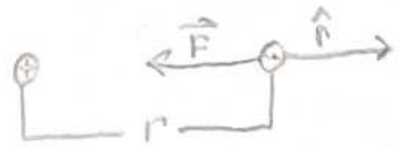
Elementary Charge: e

Proton => $+e$

Electron => $-e$

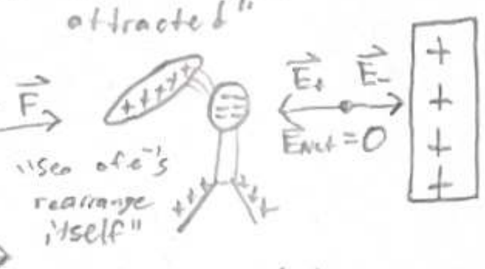
Coulomb's Law:

$\vec{F} = \frac{kq_1q_2}{r^2}$ *(Point charges only)



Charge Polarization:

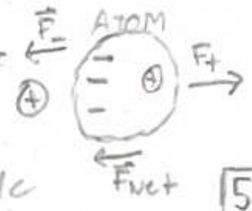
"Why are neutral objects attracted"



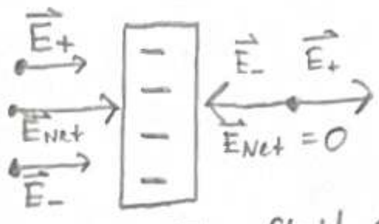
Electric Dipole:

"How can a charged comb pick up pieces of paper (insulator) if the charges inside the paper are immobile"

"Rearrange slightly within each atom"



Parallel-Plate Capacitor:



"The field strength inside the capacitor is the same at all points"

"The field outside is $\vec{0}$"

$\vec{E}_{capacitor} = \begin{cases} \frac{Q}{\epsilon_0 A}; & \text{Inside} \\ \vec{0}; & \text{Outside} \end{cases}$

where A is the surface area of each electrode

Q is the charge on each electrode (Panel)

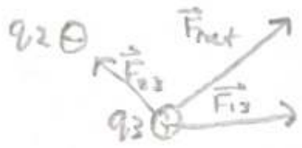
$\vec{E}_{Dipole} = \frac{\vec{p}}{4\pi\epsilon_0 r^3}$

Electric Field (Point Charge):

$\vec{E} = k\cdot\frac{q}{r^2}\hat{r}$

$k = \frac{1}{4\pi\epsilon_0}$

Superposition:

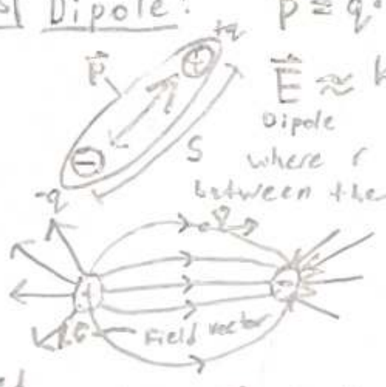


4 | Electric Fields

$\vec{E}_{NET} = \sum_{i=1}^{n}\vec{E}_i$

"Add up the fields produced by each charged object individualy"

Dipole:



$\vec{p} = q\cdot S$

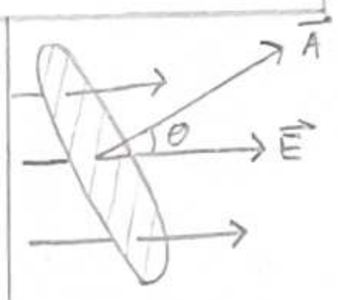
$\vec{E}_{Dipole} \approx k\cdot\frac{2\vec{p}}{r^3}$

where r is halfway between the charges

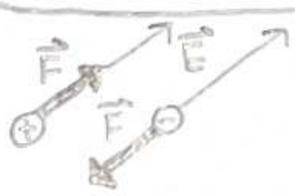
Dipole moment $= \|\vec{p}\|$

$\tau = \vec{E}q(S)\sin\theta$

"Torque maxed when $\theta = 90°$"



Motion of a charged particle in an electric field:



$\vec{F} = q\cdot\vec{E}$

- The force on a positive q is in the direction of $\vec{E}$
- The force on a negative q is in the opposite direction of $\vec{E}$

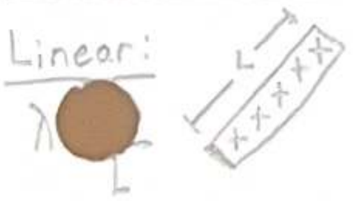
If $\vec{F}$ is the only force on q it will accelerate by

$\vec{a} = \frac{\vec{F}}{m} = \frac{q}{m}\cdot\vec{E}$

where m is the mass

Charge Density:

Linear:

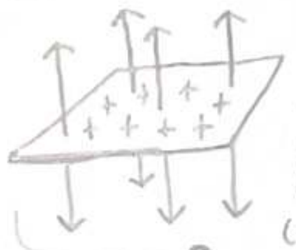


where Q = Total charge
λ = Linear Charge density

For very small lengths...

$Q = \lambda\cdot L$

Surface:



$\eta = \frac{Q}{A}$

where η = Surface Charge density per square metre (A)

(small) $Q = \eta\Delta A$

Volume:

$\rho = \frac{Q}{V}$

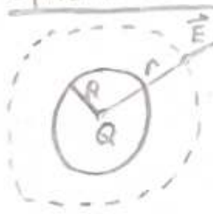
(small) $Q = \rho V$

Point Charge:

$\vec{E} = \frac{kQ}{r^2}$

$\Phi = \frac{Q}{\epsilon_0}$

Sphere of Uniform Charge:



$\Phi = \frac{Q}{\epsilon_0}$

for $r > R$

$\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2}$

for $r < R$

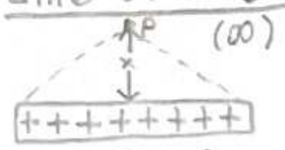
$\vec{E} = \frac{Qr}{4\pi\epsilon_0 R^3}$

Conducting:

$r > R$; Same

$r < R$; $\vec{E} = \vec{0}$

Line of Charge:

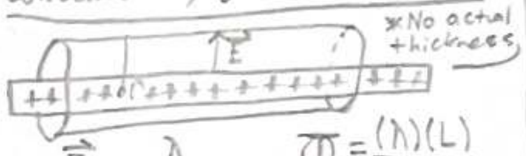


$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 x}$

Ring of Charge:

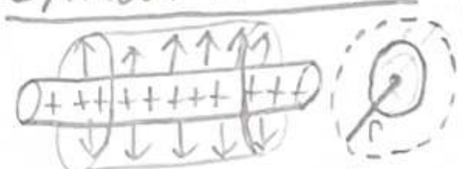
$\vec{E} = \frac{(k)(\lambda)(2\pi)(R)(x)}{(x^2+R^2)^{3/2}}$

Infinite line of charge Surrounded by gaussian Cylinder:



$\vec{E} = \frac{\lambda}{(2\pi)(r)(\epsilon_0)}$; $\Phi = \frac{(\lambda)(L)}{\epsilon_0}$

Cylinder of Charge:



$\Phi = \frac{\lambda L}{\epsilon_0}$; $\vec{E} = \frac{\lambda}{(2\pi)(\epsilon_0)(r)}$

Conducting:

$r \geq R$ (Outside) "Same $\vec{E}$ field"

$r \leq R$; $\vec{E} = \vec{0}$

Inside a Cylinder:

$\Phi = \frac{\lambda L r^2}{\epsilon_0 R^2}$

$\vec{E} = \frac{\lambda r}{2\pi\epsilon_0 R^2}$

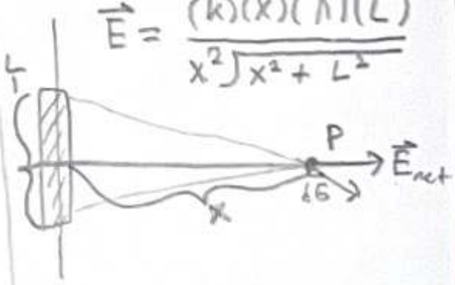
Sheet of charge:

$\Phi = \frac{\sigma A}{\epsilon_0}$ $\vec{E} = \frac{\sigma}{2\epsilon_0}$

Parallel plate: $\vec{E} = \frac{\sigma}{2\epsilon_0}$

Finite Line of Charge:

$\vec{E} = \frac{(k)(x)(\lambda)(L)}{x^2\sqrt{x^2+L^2}}$



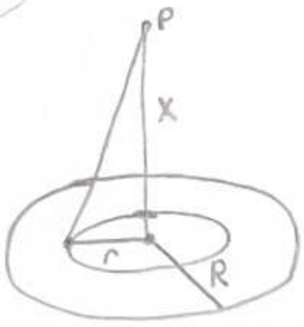
Inside a sphere of Charge:

$\Phi = \vec{E}4\pi r^2$

$\Phi = \frac{Qr^3}{\epsilon_0 R^3}$

$\vec{E} = \frac{Qr}{4\pi\epsilon_0 R^3}$

Disk of Charge:

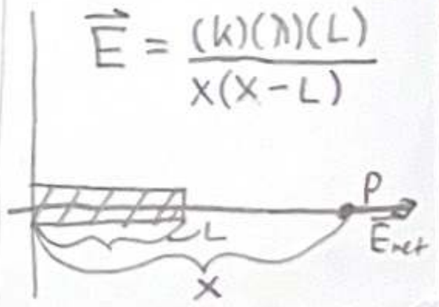


$\vec{E} = (k)(\sigma)(2\pi)\left[1 - \frac{x}{\sqrt{x^2+R^2}}\right]$

*Area charge density

Line charge directly above:

$\vec{E} = \frac{(k)(\lambda)(L)}{x(x-L)}$

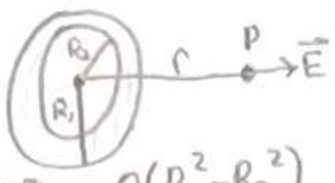


Non-Conducting Slab:

Outside: $\vec{E} = \frac{\rho d}{\epsilon_0}$

Inside: $\vec{E} = \frac{\rho z}{\epsilon_0}$

Uniform-Charge Pipe:



$\vec{E} = \frac{\rho(R_1^2 - R_2^2)}{2\epsilon_0 r}$

Prefixes:

$P = 10^{15}$	$c = 10^{-2}$
$T = 10^{12}$	$m = 10^{-3}$
$G = 10^{9}$	$\mu = 10^{-6}$
$M = 10^{6}$	$n = 10^{-9}$
$K = 10^{3}$	$p = 10^{-12}$
$h = 10^{2}$	$f = 10^{-15}$

Surface Area/Volume:

Sphere:

Area $= 4\pi r^2$

Volume $= \frac{4}{3}\pi r^3$

Cone:

Area $= \pi r s + \pi r^2$

Volume $= \frac{1}{3}\pi r^2 h$

Cylinder:

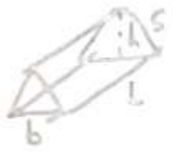
Area $= 2\pi r^2 + 2\pi r h$

Volume $= \pi r^2 h$

Prism:

Area $= bh + 2ls + lb$

Volume $= \frac{1}{2}(bh)L$



Ellipse:

Surface area $= \pi a b$

Proofs

$(\lambda)(2\pi r) = Q_{total}$

Ring of Charge:

$\vec{E} = \int_Q \frac{k\,dq}{r^2}$ $\quad * E\cos\theta$

$= \int_Q \frac{k\lambda\,ds}{(R^2+x^2)}$ $\quad * = E\left(\frac{adr}{hyp}\right)$

$= \frac{k\lambda}{(x^2+R^2)}\int_Q \frac{x}{\sqrt{R^2+x^2}}\,ds$

$= \frac{(k)(\lambda)(2\pi R)(x)}{(R^2+x^2)^{3/2}}$

Inside a Solid Insulating Sphere:

$\oint_S \vec{E}\cdot d\vec{A} = \frac{q_{net}}{\epsilon_0}$

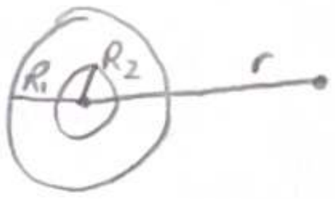
$\oint_S \vec{E}\cdot d\vec{A}\cos\theta = \frac{Qr^3/a^3}{\epsilon_0}$

$E\oint_S d\vec{A} = \frac{Qr^3}{\epsilon_0 a^3}$

$E(4\pi r^2) = \frac{Qr^3}{\epsilon_0 a^3}$

$\vec{E} = \frac{Qr^3}{4\pi\epsilon_0 a^3}$

Pipe: (Infinitely long)



$E = \frac{Q}{A\epsilon_0}$ $\quad V = \pi(R_2^2 - R_1^2)L$

$A = 2\pi r L$

$E = \frac{\pi\rho(R_1^2 - R_2^2)L}{2\pi L\epsilon_0} = \frac{\rho(R_1^2 - R_2^2)}{2r\epsilon_0}$

Finite Line of Charge:

$\vec{E} = \left(\frac{kq}{r^2}\right)\cos\theta$

$= \left(\frac{kq}{r^2}\right)\left(\frac{x}{\sqrt{x^2+L^2}}\right)$

$= \int_{-a}^{a}\left(\frac{k\lambda dL}{(x^2+L^2)}\right)\left(\frac{x}{\sqrt{x^2+L^2}}\right)$

$= \frac{kx\lambda L}{x^2\sqrt{x^2+L^2}}$

Infinite slab:

$EA = \frac{\rho A d}{\epsilon_0}$

$Q_{enc} = \rho \times \text{Volume} = \rho A d$

$\int_{Total}\vec{E}\cdot d\vec{A} = \left(\int_{Base}(\vec{E}\cdot d\vec{A} + \int_{laterals}\vec{E}\cdot d\vec{A} + \int_{Top}\vec{E}\cdot d\vec{A}\right.$

$\vec{E} = \frac{\rho d}{\epsilon_0}$ $\quad z \geq d$ $\qquad \vec{E} = \frac{\rho z}{\epsilon_0}$, (inside)

$\vec{E} = -\frac{\rho d}{\epsilon_0}$ $\quad z \leq d$

Outside a Sphere:

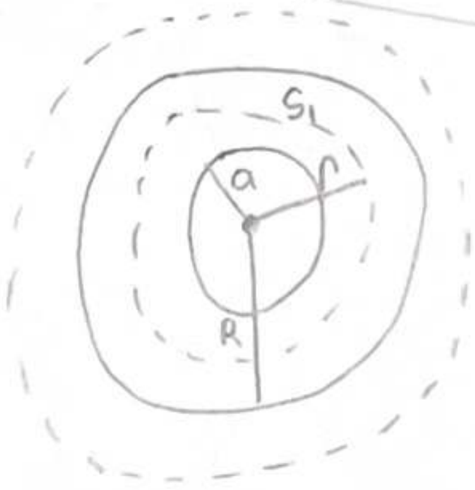
$\oint \vec{E}\cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$

$\oint \vec{E}\cdot d\vec{A}\cos\theta = \frac{Q}{\epsilon_0}$

$E(4\pi r^2) = \frac{Q}{\epsilon_0}$

$E = \frac{Q}{4\pi\epsilon_0 r^2}$

$= \frac{kQ}{r^2}$



$V_{TOT} = \frac{4}{3}\pi R^3 - \frac{4}{3}\pi a^3$

$= \frac{4}{3}\pi(R^3 - a^3)$

$V_1 = \frac{4}{3}\pi r^3 - \frac{4}{3}\pi a^3$

$= \frac{4}{3}\pi(r^3 - a^3)$

$Q\left(\frac{V_2}{V_{TOT}}\right) = q_{net}$

$\oint_{S_1}\vec{E}\cdot d\vec{A} = \frac{q_{net}}{\epsilon_0}$

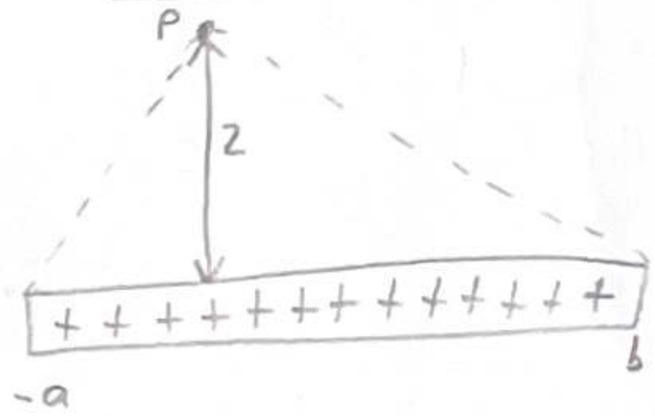
$= \oint_{S_1} E\,dA\cos\theta = \frac{1}{\epsilon_0}\left(\frac{\frac{4}{3}\pi(r^3-a^3)}{\frac{4}{3}\pi(R^3-a^3)}Q\right)$

$E\oint_{S_1} dA = \frac{Q}{\epsilon_0}\left(\frac{r^3-a^3}{R^3-a^3}\right)$

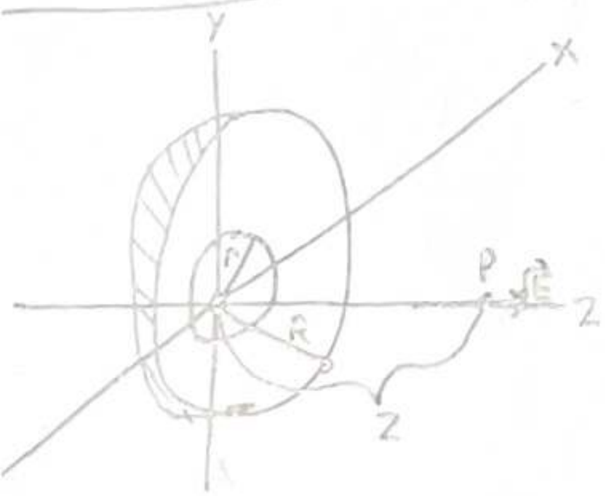
$E(4\pi r^2) = \frac{Q}{\epsilon_0}\left(\frac{r^3-a^3}{R^3-a^3}\right)$

$\vec{E} = \boxed{\frac{Q}{4\pi\epsilon_0}\cdot\frac{1}{r^2}\left(\frac{r^3-a^3}{R^3-a^3}\right)}$

Finite line of [redacted]



$E_z = \frac{k\lambda}{z}\left[\frac{b}{(z^2+b^2)^{1/2}} + \frac{a}{(z^2+a^2)^{1/2}}\right]$



We know Field due to a ring at origin is along the z-axis and $\vec{E} = \frac{kQz}{(z^2+a^2)^{3/2}}$

$\vec{E} = \int d\vec{E}$

$= \int_Q \frac{kz\,dq}{(z^2+r^2)^{3/2}}$

what is dq?

$\frac{2\pi r\,dr}{\pi R^2} = \frac{dq}{Q}$

$dq = \frac{2Qr\,dr}{R^2}$

$\vec{E} = \int_0^R \frac{kz\left(\frac{2Qr\,dr}{R^2}\right)}{(z^2+r^2)^{3/2}}$

$\vec{E} = \left(\frac{2kQz}{R^2}\right)\int_0^R \frac{r\,dr}{(z^2+r^2)^{3/2}}$

$u = z^2 + r^2$, $2(r\,dr) = du$

$\frac{du}{dr} = 2r$, $r\,dr = \frac{du}{2}$

$\vec{E} = \frac{2kQz}{R^2}\int_{u_1}^{u_2}\frac{du/2}{u^{3/2}}$

$\vec{E} = \left(\frac{kQz}{R^2}\right)\left(\frac{-1}{2}\right)u^{-1/2}\Big|_{u_1}^{u_2}$

$= -\left(\frac{kQz}{R^2}\right)\frac{1}{\sqrt{z^2+r^2}}\Big|_0^R$

$= \left(\frac{kQz}{2R^2}\right)\left(\frac{1}{\sqrt{z^2+R^2}} - \frac{1}{\sqrt{z^2}}\right)$

$\vec{E} = \left(\frac{kQz}{2R^2}\right)\left(\frac{1}{z} - \frac{1}{\sqrt{z^2+R^2}}\right)$

$\boxed{\vec{E} = \left[\left(\frac{kQ}{2R^2}\right)\left(1 - \frac{z}{\sqrt{z^2+R^2}}\right)\right]}$

$\int_{S_1}\vec{E}\cdot d\vec{A} + \cancel{\int_{S_2}\vec{E}\cdot d\vec{A}}^{\,0} + \int_{S_3}\vec{E}\cdot d\vec{A}^{\,0} = \frac{\sigma A}{\epsilon_0}$

$2\int_{S_1}\vec{E}\cdot d\vec{A}\,\cancel{\cos\theta}^{\,1} = \frac{\sigma A}{\epsilon_0}$

$2E\int_{S_1} dA = \frac{\sigma A}{\epsilon_0}$

$2EA = \frac{\sigma A}{\epsilon_0}$

$\boxed{\vec{E} = \frac{\sigma}{2\epsilon_0}}$